

Assignment #1: Storyboard
Professor Gabby Resch
Data Visualization and Storytelling
Jin Chan
2026-02-15

Introduction

In this assignment, I want to explore a concern I've been noticing more and more often: when AI is used as a shortcut, not as a tool, it may quietly reduce the practice of critical thinking, especially for younger learners.

The target audience for my storyboard is parents and after-school tutors in Canada. They are often the adults who most directly shape children's study habits outside the classroom. I hope this data story will help them recognize early warning signs and adopt practical ways to encourage critical thinking when children use AI tools.

As I talked to you earlier, I am currently working for a company called Code Ninjas, where I teach kids how to use different platforms to make games.

I've been doing this for about two and a half years, and I've met students with a wide range of learning styles. Some pick things up quickly, some need more time, and some need a different kind of support to stay engaged.

I genuinely enjoy working with kids, and I can tolerate kids with different personalities. I've taught students who are energetic, stubborn, easily distracted, or extremely shy. And I even worked with kids that have ADHD and autism before. I think they are cute in their own way and I enjoy sharing my knowledge with them. But recently, one pattern has been testing my patience more than any personality difference: some students are struggling to slow down, read carefully, and reason through a problem step by step.

I'm from the 1997 generation, so I grew up watching technology move fast. I always like to learn new things, like memes, brain rots, new technology, and exactly AI related things. As a part time game developer, I love AI. It helps me with brain storming, art and programming. It helps me brainstorm, iterate on code, and speed up debugging. It helps me to save time on boring tasks like programming, and debugging, so I can focus on the things I am interested in like game design and creative work.

Without a doubt, AI is clearly a powerful innovation. But I keep thinking about something my elementary school math teacher once told me: "The invention of calculators makes basic arithmetic faster, so mathematicians can work on harder math problems. But to know how to use a calculator, you must know how to do the basic math"

Back then, I didn't think much about it. But now, when I am working as a teacher, I start to share the same view as she did.

Here is a conversation I have been having more and more often:

Student: What is this error?

Me: If you need my help, you need to tell me more. I am not a psyker, what did the console tell you?

Student: It says null reference exception.

Me: Okay. So, what is a null reference exception?

Student: I don't know. How would I know?

Me: How about we paste the error message in ChatGPT and see what it tells you?

Student: (Paste the Whole Script in ChatGPT) Um... I don't know. It said drag the Game Object into the variable in the inspector.

Me: (Start to feel headache) It is the window on the right side.

Student: I don't know. How to drag the Game Object into the variable.

Me:

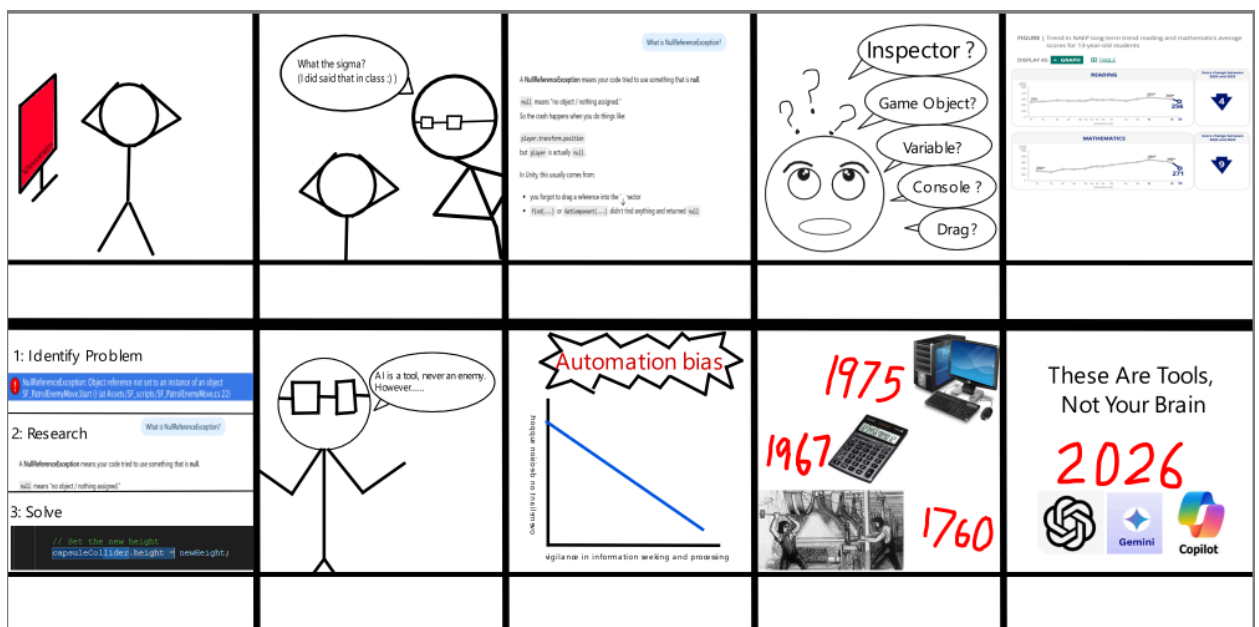
Something has changed: the problem is no longer only "they don't know the answer," but "they don't know how to look for the answer." That shift is what I want to examine with data. I want to know what is going on with the new generation. It may reveal how tool use is shaping the habits behind learning.

Therefore, in this assignment, I will use scrollytelling to tell a data story based on what I see in my daily work at Code Ninjas, and to explore how AI tools may be changing students' learning habits.

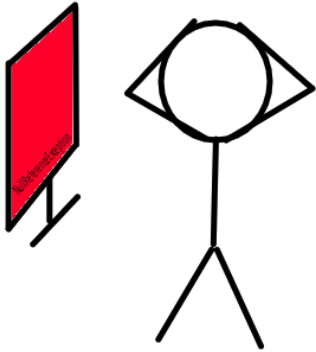
In my first iteration, the first idea in my mind is to present the story as a reality-show style video based on my day to day work at Code Ninjas. The tone would

be similar to Hell's Kitchen: fast-paced, humorous, and slightly exaggerated to highlight the issue in an entertaining way. However, I only had three days left when I started writing this assignment, and I did not have enough time to request permission to film inside the center. Because of this constraint, I decided to shut down the video concept and explore a format that could be produced more quickly.

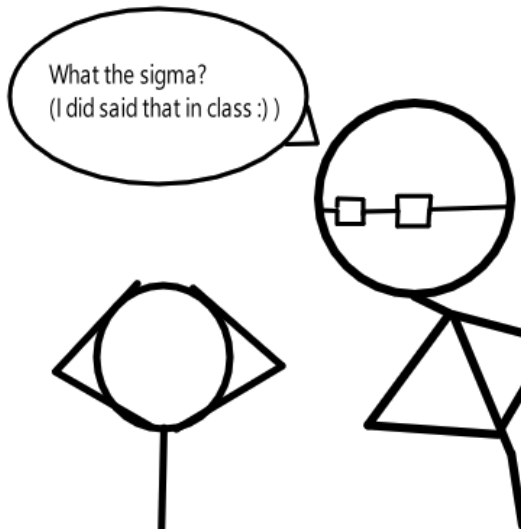
In my second iteration, I used a software called Krita to sketch out my ideas and began searching for data to support my story. During this stage, my biggest discovery was that the scenario I described earlier actually has a name: “automation bias.” It refers to the tendency for users to become overly reliant on decision-support tools, to the point that they reduce their vigilance in information seeking and processing.



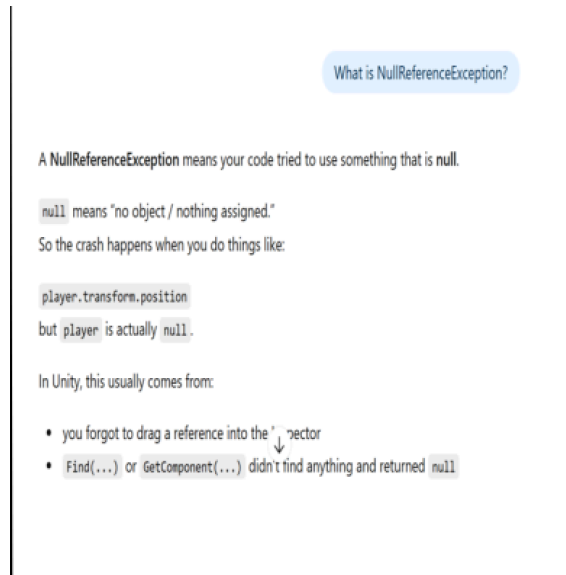
In the final iteration, I reorganized the image sequence and inserted additional explanatory text between panels to create a smoother, scrollytelling-like flow. Below is the final storyboard design.



Scene#1: A close-up of the classroom shows a student looking up and asking, "What is this error?" A small Unity Console icon is displayed nearby.



Scene#2: The conversation continues, You ask, "What the sigma?"and the student says "NullReferenceException." The picture pauses on the student's blank expression.



Scene#3: I told the student to search up "NullReferenceException." , but instead, the student pasted the **entire script** into the ChatGPT, and the screen displayed the AI's response in bubbles.



Scene 4: A crucial turning point. Students don't know where the Inspector is or how to drag it. The narration points out the changes you observe; the problem isn't that they can't write, but that they don't know how to find clues and verify them.

FIGURE | Trend in NAEP long-term trend reading and mathematics average scores for 13-year-old students

DISPLAY AS **GRAPH** TABLE



Scene 5:

A graph from The Nations Reportcard shows that 13-year-old students have experienced a decline in reading and math

<https://www.nationsreportcard.gov/highlights/ltr/2023/>

1: Identify Problem

! NullReferenceException: Object reference not set to an instance of an object
SF_PatrolEnemyMove.Start () (at Assets/SF_scripts/SF_PatrolEnemyMove.cs:22)

2: Research

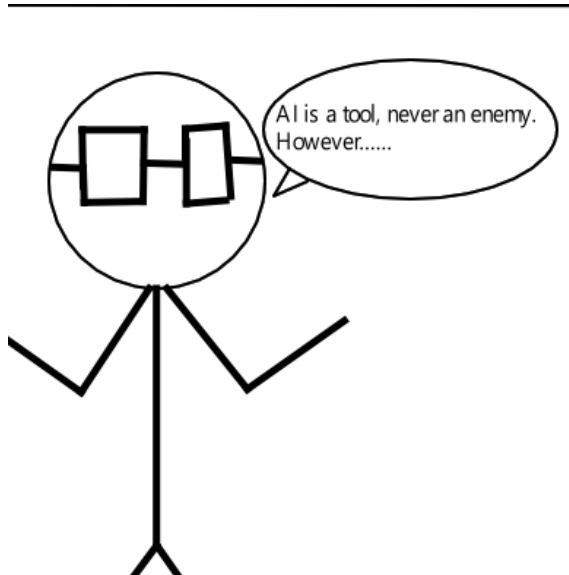
What is NullReferenceException?

A NullReferenceException means your code tried to use something that is null.

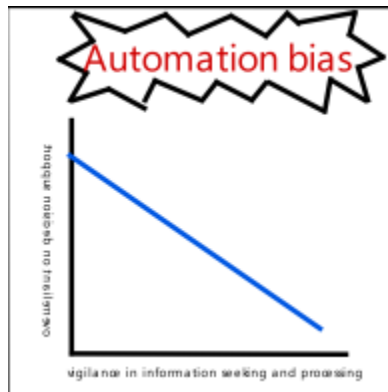
null means "no object / nothing assigned."

3: Solve

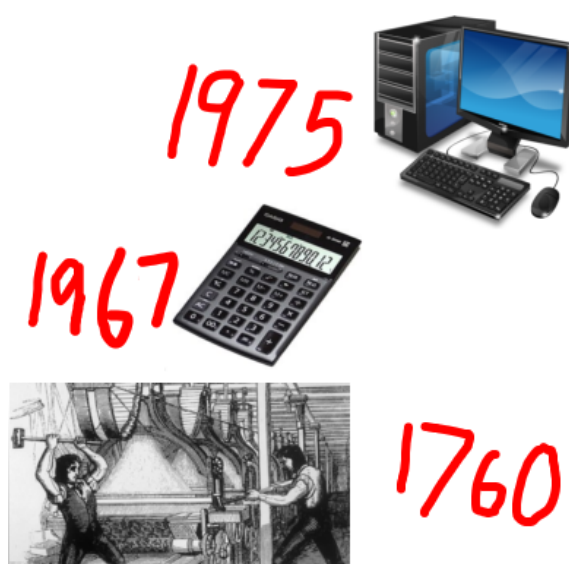
```
// Set the new height  
capsuleCollider.height = newHeight;
```



Scene 6/7: Example how I used to solve my error.

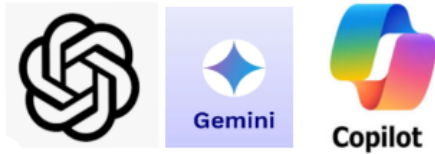


Scene 8: Explain Automation Bias.



These Are Tools,
Not Your Brain

2026



Scene 8/9: Show off different tools that was invented to improve human performance.

Example: 1760 Industrial Revolution, 1967 First modern Electronic Handheld Calculator,

1975 First commercially successful personal computer.